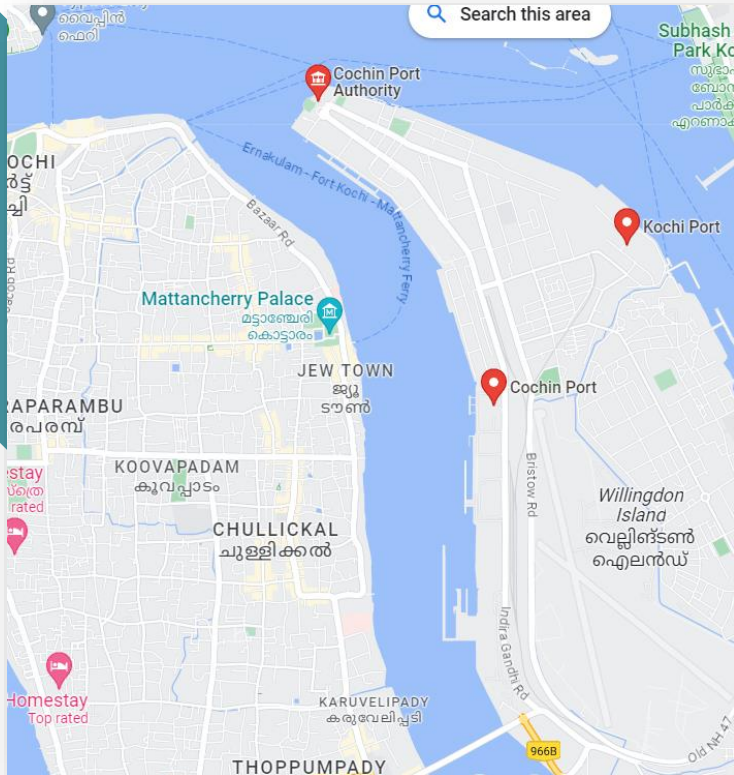




**SMART INDIA
HACKATHON
2023**

Basic Details of the Team and Problem Statement

COCHIN PORT MAP VIEW



Ministry : Ministry of Ports, Shipping and Waterways

PS Code: SIH1310

Problem Statement Title: Drone-based surveillance system for the vessels plying in port areas and encroachments

Team Name: PowerX

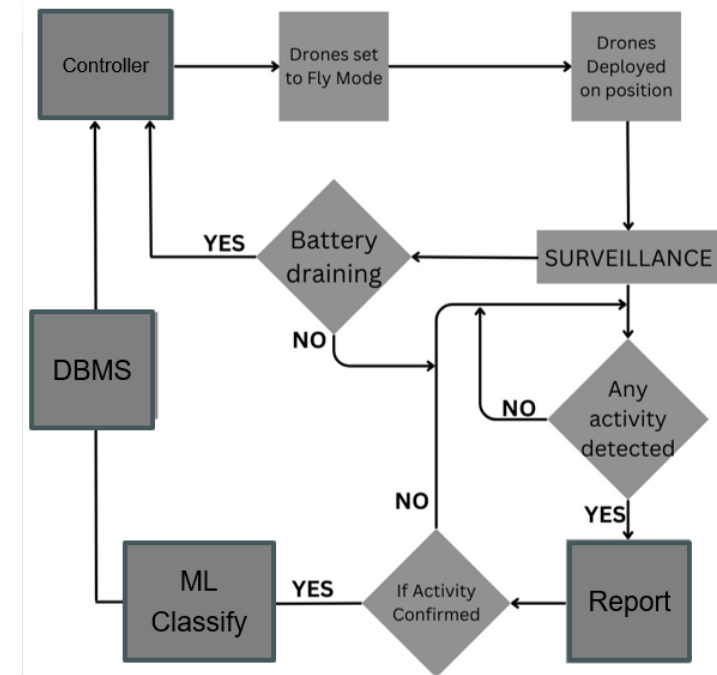
Team Leader Name: Mawwiz Shaikh

Theme Name: Robotics and Drone

Idea/Approach Details

Our idea/Solution here:

- In the current work, we propose system that is capable of:
- 1. Keeping track of all the boats and similar vehicles in the Port's waters.
- 2. System that gives info about encroachments.
- 3. Realtime Data and Video streaming.
- 4. Drone has optimized machine learning approach to detect intruders.
- 5. Reported encroachments will have a database management system.



Describe your Technology stack here:

- Motors
- Battery
- GPS
- Camera
- Arduino
- RF
- App/Web

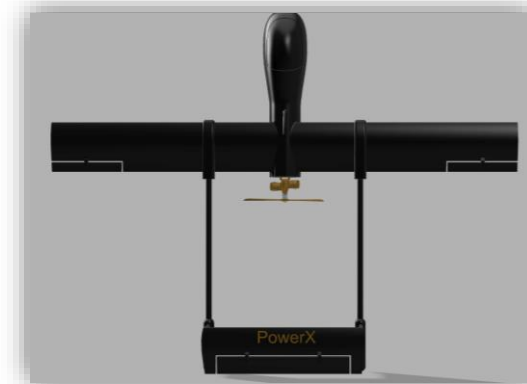
Idea/Approach Details

❑ Use Cases

- Continuous Surveillance
- Real-Time Data Transmission
- Automated Response
- Enhanced Situational Awareness
- Resource Optimization

❑ Show stopper

- Centralized system
- Autonomous Drone Capability
- Long-Range Battery Technology
- Obstacle Avoidance
- Multi-Sensor Fusion
- Remote Charging and Deployment Stations
- Adaptive Detection
- Machine Learning for Detection



Team Member Details

Team Leader Name: Mawwiz Shaikh

Branch : Btech

Stream : CSE

Year : III

Team Member 1 Name: Sanket Pawar

Branch : Btech

Stream : CSE

Year : III

Team Member 2 Name: Deepanshu Sharma

Branch : Btech

Stream : CSE

Year : III

Team Member 3 Name: Gaus Sayyad

Branch : Btech

Stream : ECE

Year : III

Team Member 4 Name: Karankumar Nevage

Branch : Btech

Stream : ECE

Year : III

Team Member 5 Name: Apurva Pathak

Branch : Btech

Stream : ECE

Year : II